PATTERN_ST_MARKOV: |ST|² label chain → T-fibre vs MF2

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RESULTS: PATTERN_ST_MARKOV (Map-PST1, Map-PST2)

PREREG sha256 2b592b5af2b3a07877e0c5565d460b7945f930fc18c5b752808b574f4c7fb4ca. Hashed after explore and before test.

Class structure of |ST|² (and of |TS|², identical): doubly stochastic, 2 closed classes, each with a uniform stationary law. - C105: coset types (0,0),(0,1),(1,0); 105 labels. - C35: type (1,1); 35 labels. Compared with |S|² (PSM1, 3 classes), T merges (0,0) with the 2-cycle.

fibre	ρ_res (MF2)	PST1 (uniform 140)	PST2 (C105)	C35
0	1/32	8/35	32/105	0
1	1/4	11/70	11/105	11/35
2	3/32	0	0	0
3	0	9/70	3/35	9/35
4	7/32	19/70	38/105	0
5	1/4	3/35	2/35	6/35
6	5/32	0	0	0
7	0	9/70	3/35	9/35

- PST1: TV = 71/140. **FAIL.**
- PST2: TV = 247/420. **FAIL.**
- No stationary mix reaches fibres 2 or 6. Category A. Bridge: **NO.** ## NOTES: PATTERN_ST_MARKOV ## Lemma PST1 (Category A; closed obstruction) The T-phase fibre map labels → Z/8 has image {0,1,3,4,5,7}: **no label** has T-fibre 2 or 6 (PB2 multiset). Therefore **every** probability law on the 140 labels pushes to zero mass on {2,6}. That covers any stationary law of any chain built from the written S and T (|S|², |ST|², |TS|², or any word in S,T), and any weighting at all. MF2 has mass 3/32 + 5/32 = 1/4 on {2,6}. So **no written modular 8-fibre law from S/T label dynamics under the T-phase fibre can equal MF2.** PSM1(b) and PST2 are corollaries.

Lemma PST2 (Category A, structural)

$|ST|^2$ has exactly two closed classes, of sizes 105 and 35. The $105 = 35 \cdot 3$ split matches the written count of 105 companions. That is an observation about the label dynamics; it is not used as a bridge.

Closed obstruction summary (written Resonant 8-fibre laws vs MF2)

Lemma	Fibre coordinate	Result
BATCH8, PF1	square thetas, unshifted or label-shifted	≤3 classes
PQ1 / PS1	140Q fibre, any weights	misses {1,5,6}
PB1	B(ν,a)	uniform
PB2 / PSM1 / PST1	T-phase fibre, any law on labels incl. all S/T dynamics	misses {2,6}

Every written fibre coordinate on the Resonant side is closed against MF2.

Next step (per stop rule)

Report to Arvin. A further map needs a Resonant-side 8-fibre coordinate or grading Π that is not in the written sources. It must not be invented here.